

Distality in ordered abelian groups

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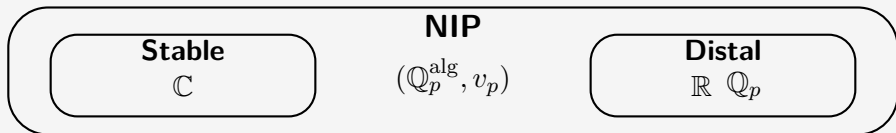
Model Theory Summer School
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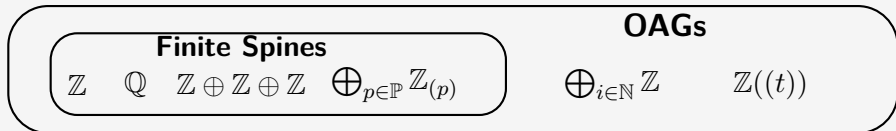


Summary

Distality is the opposite of stability (among NIP theories).



Though all ordered abelian groups (OAGs) have NIP [GS84], only a distality criterion for OAGs with finite spines was known [Asc+22].



We have

- given a distality criterion for *all* OAGs.
- shown all OAGs have distal expansions.

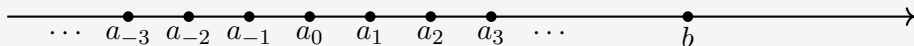
Definition ([Sim13])

A theory is *distal* if for any sequence $(\bar{a}_i)_{i \in \mathbb{Q}}$ and any tuple $\bar{b}$, if

- $(\bar{a}_i)_{i \in \mathbb{Q}}$ is indiscernible and
- $(\bar{a}_i)_{i \in \mathbb{Q} \setminus \{0\}}$ is indiscernible over $\bar{b}$,

then $(\bar{a}_i)_{i \in \mathbb{Q}}$ is indiscernible over $\bar{b}$.

In DLO...



Roughly speaking, distal theories are those NIP theories which do not have a stable part.

Example (Distal theories)

- DLO
- $(\mathbb{R}; +, \cdot), (\mathbb{Q}_p; +, \cdot)$
- $(\mathbb{Z}; <, +)$

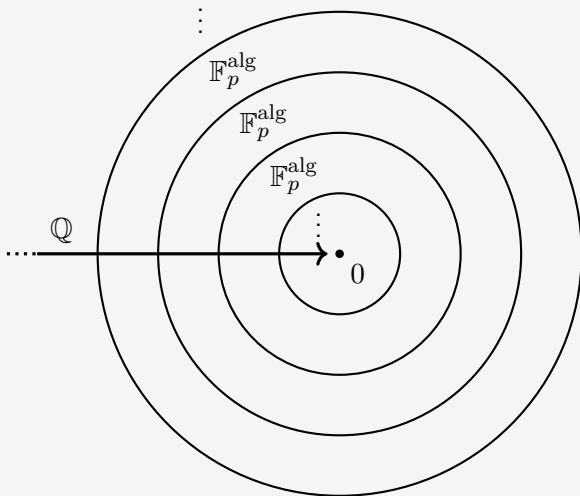
Example (NIP non-distal theories)

- Stable theories (with infinite models)
- $(\mathbb{Q}_{>0}; <, \cdot)$ ($|\mathbb{Q}_{>0}/(\mathbb{Q}_{>0})^p| = \infty$ for any prime p)
- $(\mathbb{Q}_p^{\text{alg}}; +, \cdot, v_p)$

Example (Theories without a distal expansion)

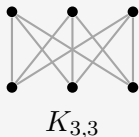
- Theories interpreting an infinite field of characteristic $p > 0$ ([CS18, Corollary 6.3]).

A valued field $(\mathbb{Q}_p^{\text{alg}}; +, \cdot, v_p)$ is a combination of its value group $\mathbb{Q}$ and residue field $\mathbb{F}_p^{\text{alg}}$.



Why distality? —Zarankiewicz's problem

Fix $k \geq 2$. Let $(V_1, V_2; E)$ be a $K_{k,k}$ -free bipartite graph and $n = \max\{|V_1|, |V_2|\}$. It is well-known ([KST54]) that



$$|E| = O(n^{2-(1/k)}).$$

If V_1 is points in $\mathbb{R}^2$, V_2 is lines in $\mathbb{R}^2$, and E is the incidence relation (this graph is $K_{2,2}$ -free), it is known ([ST83]) that

$$|E| = O(n^{4/3}).$$

Chernikov, Galvin, and Starchenko [CGS20] showed that in a distal theory, for every formula $\varphi(\bar{x}, \bar{y})$ there are $t, d \in \mathbb{N}_{\geq 2}$ such that

$$\text{if } (V_1, V_2; \varphi(V_1, V_2)) \text{ is } K_{k,k}\text{-free, then } |\varphi(V_1, V_2)| = O(n^{\frac{2td-t-d}{td-1}}).$$

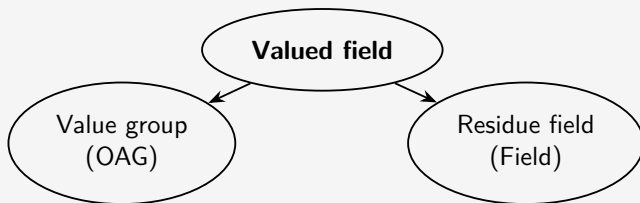
Why OAGs? —AKE principle

We need to study OAGs to study valued fields:

Theorem (Aschenbrenner, Chernikov, Gehret, and Ziegler [Asc+22])

Let (K, v) be a henselian valued fields. (K, v) is distal (has a distal expansion) iff

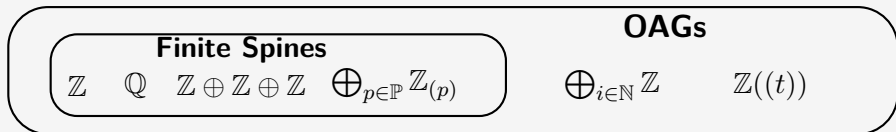
- *(K, v) is finitely ramified,*
- *The value group of K is distal (has a distal expansion) as an OAG, and*
- *The residue field of K is distal (has a distal expansion) as a pure field.*



However, only a partial result was known for OAGs. Fix an OAG G :

Theorem (ibid.)

If G has finite 'spines', then G is distal iff $|G/pG| < \infty$ for all primes p .



Theorem (O. [Oku26])

G is distal iff for every prime p , the sizes of the 'ribs' with respect to a valuation $\mathfrak{s}_p$ is uniformly bounded.

Theorem (O. [Oku26])

Every OAG has a distal expansion.

To study the theory of an OAG G , it is essential to consider definable convex subgroups of G .

Definition

For $n \geq 2$ and $a \in G \setminus nG$, let $\mathfrak{s}_n(a)$ be the largest convex subgroup H such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$.

Lemma

$\mathfrak{s}_n(a)$ is definable.

Proof.

Let $\langle x \rangle^{\text{conv}}$ be the convex subgroup generated by x . Then, $x \in \mathfrak{s}_n(a) \iff \langle x \rangle^{\text{conv}} \subseteq \mathfrak{s}_n(a) \iff a \notin \langle x \rangle^{\text{conv}} + nG = [0, nx] + nG$. ■

For convex subgroups H_1, H_2 , we write $H_1 \leq H_2 \iff H_1 \subseteq H_2$.

Example

Let $G = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ with the reversed lexicographic order. The convex subgroups are $(0) \oplus (0) \oplus (0) < \mathbb{Z} \oplus (0) \oplus (0) < \mathbb{Z} \oplus \mathbb{Z} \oplus (0) < G$. For $a = (0, 1, 0)$, $a \notin \mathbb{Z} \oplus (0) \oplus (0) + 2G$ and $a \in \mathbb{Z} \oplus \mathbb{Z} \oplus (0) + 2G$. Hence, $\mathfrak{s}_2(a) = \mathbb{Z} \oplus (0) \oplus (0)$.

An important fact is that we may regard $\mathfrak{s}_n$ as a valuation on G :

Lemma

- 1 $\mathfrak{s}_n(-a) = \mathfrak{s}_n(a)$.
- 2 $\mathfrak{s}_n(a + b) \leq \max \{\mathfrak{s}_n(a), \mathfrak{s}_n(b)\}$ (*ultrametric inequality*).

Proof of (2).

For a convex subgroup H , if $H > \max \{\mathfrak{s}_n(a), \mathfrak{s}_n(b)\}$ then $a, b \in H + nG$, and hence $a + b \in H + nG$. Thus $H > \mathfrak{s}_n(a + b)$. ■

We view $(G, \mathfrak{s}_n)$ in an analogue to valued fields.

Valued field	OAG
v	$(\mathfrak{s}_n)_{n \geq 2}$
Value group	spines
Residue field	ribs

Definition

We define the *spine* $\mathcal{S}_n = \text{Im}(\mathfrak{s}_n)$.

For $G = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$,

$$\mathcal{S}_2 = \{\emptyset < (0) \oplus (0) \oplus (0) < \mathbb{Z} \oplus (0) \oplus (0) < \mathbb{Z} \oplus \mathbb{Z} \oplus (0)\}.$$

Theorem (Aschenbrenner, Chernikov, Gehret, and Ziegler [Asc+22])

Suppose that $|\mathcal{S}_p| < \infty$ for every prime p . Then G is distal iff $|G/pG| < \infty$ for all primes p .

For $G = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$, $G/pG \cong (\mathbb{Z}/p\mathbb{Z}) \oplus (\mathbb{Z}/p\mathbb{Z}) \oplus (\mathbb{Z}/p\mathbb{Z})$. Hence it is distal.

Definition

For $H \in \mathcal{S}_n \setminus \{\emptyset\}$, let $G_{\leq H}^{s_n} = \{x \in G \mid s_n(x) \leq H\}$ and $G_{< H}^{s_n}$ (these are subgroups of G). We define $R_H^{s_n} = G_{\leq H}^{s_n} / G_{< H}^{s_n}$, the *rib* at H .

For $G = \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ and $H = \mathbb{Z} \oplus (0) \oplus (0)$, $G_{\leq H}^{s_2} = \mathbb{Z} \oplus \mathbb{Z} \oplus 2\mathbb{Z}$ and $G_{< H}^{s_2} = \mathbb{Z} \oplus 2\mathbb{Z} \oplus 2\mathbb{Z}$. Thus $R_H^{s_2} \cong \mathbb{Z}/2\mathbb{Z}$.

Lemma

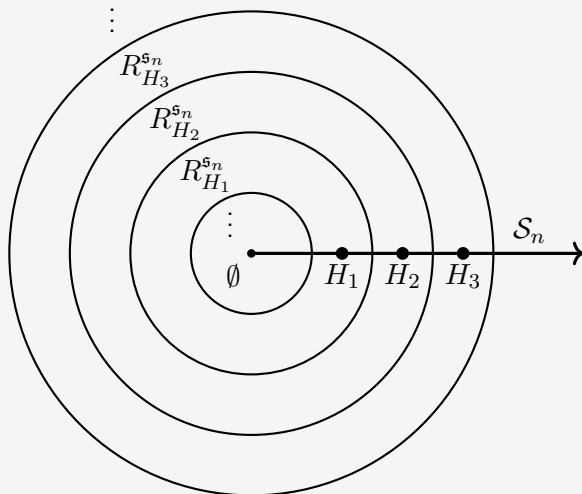
The rib $R_H^{s_n}$ is of exponent n : For every $a + G_{< H}^{s_n} \in R_H^{s_n}$, $n(a + G_{< H}^{s_n}) = 0$.

Proof.

Since $s_n(na) = \emptyset$, we have $na \in G_{< H}^{s_n}$ and hence $n(a + G_{< H}^{s_n}) = 0$. ■

In particular, $R_H^{s_p}$ is an $\mathbb{F}_p$ -vector space.

Valued field	OAG
v	$(s_n)_{n \geq 2}$
Value group	spines
Residue field	ribs



Theorem (O. [Oku26])

G is distal iff for every prime p , there is $N_p \in \mathbb{N}$ such that $|R_H^{s_p}| < N_p$ for every $H \in \mathcal{S}_p \setminus \{\emptyset\}$.

If G is sufficiently saturated, G is distal iff for every prime p , $|R_H^{s_p}| < \infty$ for every $H \in \mathcal{S}_p \setminus \{\emptyset\}$.

$R_H^{s_p}$ is a $\mathbb{F}_p$ -vector space (hence stable). So, the distality of an OAG is determined by whether or not an infinite vector space appears as a rib.

While some OAGs are not distal, we have also shown:

Theorem (O.)

Every OAG has a distal expansion.

An infinite rib makes an OAG non-distal. Hence, our strategy is to introduce another valuation on each rib so that resulting ribs are finite. Recall that $R_H^{\mathbb{F}_p}$ is a $\mathbb{F}_p$ -vector space.

Example

Let $V = \bigoplus_{i \in I} \mathbb{F}_p$ be a $\mathbb{F}_p$ -vector space. Introduce a linear order on the index set I and define a valuation v on V as follows:
 $v((a_i)_{i \in I}) = \max \{i \in I \mid a_i \neq 0\}$. Then, every rib with respect to v is isomorphic to $\mathbb{F}_p$. Furthermore, $(V, +, v)$ is distal.

Tools for Proof

OAGs have other families of valuations:

Definition

For $n \geq 2$ and $a \in G$, let $\mathfrak{t}_n(a) = \bigcup_{H \in \mathcal{S}_n, a \notin H} H$ and $\mathfrak{t}_n^+(a) = \bigcap_{H \in \mathcal{S}_n, a \in H} H$. Let $\mathcal{T}_n = \text{Im}(\mathfrak{t}_n)$ and $\mathcal{T}_n^+ = \text{Im}(\mathfrak{t}_n^+)$.

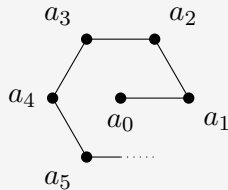
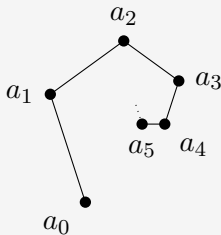
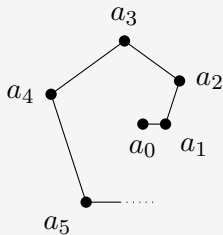
Cluckers and Halupczok [CH11] introduced a many-sorted language $(G, (\mathcal{S}_p), (\mathcal{T}_p), (\mathcal{T}_p^+); (\mathfrak{s}_{p^r}), (\mathfrak{t}_p))$ (p is prime, $r \geq 1$), and established home-sort quantifier elimination.

Distality involves indiscernible sequences. Values of indiscernible sequences behave well.

Lemma

Let $(a_i)_{i \in I}$ be an indiscernible sequence, and let $v \in \{\mathfrak{s}_n, \mathfrak{t}_n \mid n \geq 2\}$. Then exactly one of the following holds:

- $v(a_j - a_i) < v(a_k - a_j) = v(a_i - a_k)$ for every $i < j < k$ in I .
- $v(a_k - a_j) < v(a_i - a_k) = v(a_j - a_i)$ for every $i < j < k$ in I .
- $v(a_j - a_i) = v(a_k - a_j) = v(a_i - a_k)$ for every $i < j < k$ in I .

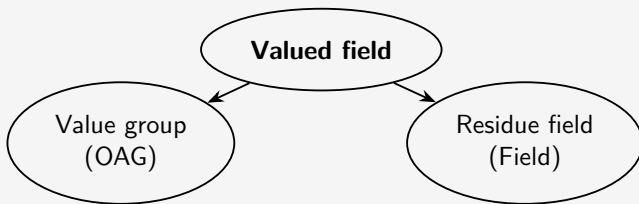


Typically, *AKE-principles* are theorems of the following form:

AKE-Principle

Suppose that a henselian valued field has some kind of tameness (such as, equicharacteristic 0). Then, it has a property P iff both its value group and residue field have P .

Examples of P : elementary equivalence, NIP, dp-rank, NTP_2 , burden, NATP, distality, distal expansions, elimination of imaginaries, stable embeddedness, etc.

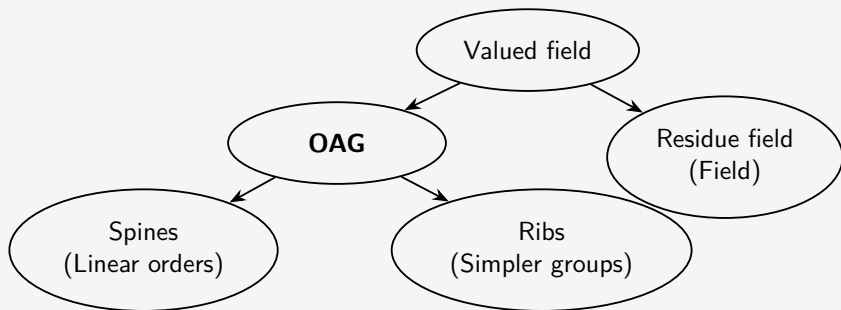


Since OAGs themselves have valuations, we may ask:

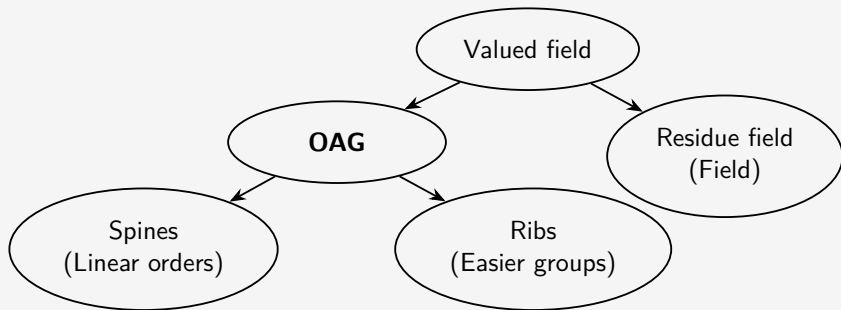
Question

What kind of AKE-principles can we establish for OAGs?

Some have already been established: elementary equivalence [Sch82, Corollary 4.7], NIP [GS84], dp-rank [Far17, Theorem 7.2], and a partial result of stable embeddedness [HLT23].



Our results may be viewed as AKE-principles of OAGs for distality. Every spine $\mathcal{S}_p$ is a linearly ordered set, which is distal. As for ribs, a vector space is distal iff it is finite. Thus, a sufficiently saturated OAG is distal iff for every prime p and $H \in \mathcal{S}_p \setminus \{\emptyset\}$, the rib at H is distal. Also, every $\mathbb{F}_p$ -vector space has a distal expansion, and so every OAG does.



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